CST402 DISTRIBUTED COMPUTING

MODULE 1			
Sl. No	**Questions**	**Marks**	**KTU/KU Month/Year**
1	List any 4 issues in the design of a distributed system.	4	DEC 18, Sep-20
2	Identify any three distributed applications and for each application, determine which all motivating factors are important for building an application over a distributed system.	3	MODEL QUESTION PAPER
3	Write notes on mobile and ubiquitous computing.	4	DEC 18
4	Assume that the surface of the past cone form a consistent cut. Does it mean that all events on the surface of the past cone are always concurrent? Demonstrate with the help of an example.	3	MODEL QUESTION PAPER
5	Explain how multimedia services are supported in distributed systems.	4	MAY19
6	Which are the different versions of send and receive primitives for distributed communication? Explain.	8	MODEL QUESTION PAPER
7	"The absence of these two transparencies most strongly affects the utilization of distributed resources". Identify and explain the above two types of transparencies with examples.	4	MAR 2020
8	Explain the three different models of the service provided by communication networks.	6	MODEL QUESTION PAPER
9	Explain how the causal dependency between events in distributed execution is defined using Lamport's happened before relationship.	8	MODEL QUESTION PAPER
10	Address the various strategies that can be adopted to satisfy the requirements of a reliable and fault tolerant distributed system.	6	MODEL QUESTION PAPER
11	In what all aspects distributed systems are better than centralized systems? Give examples of two applications for which distributed systems will be more suitable.	4	SEP 2020
12	What are the different communicating entities in a distributed system?	4	SEP 2020

MODULE 2			
Sl. No	**Questions**	**Marks**	**KTU/KU Month/Year**
1	Illustrate bully algorithm for electing a new leader. Does the algorithm meet liveness and safety conditions?	7	MODEL QUESTION
2	Clearly mentioning assumptions, explain the rules of termination detection using distributed snapshots.	7	MODEL QUESTION
3	In Chandy-Lamport algorithm for recording global snapshots, explain how the recorded local snapshots can be put together to create the global snapshot. Can multiple processes initiate the algorithm concurrently?	7	MODEL QUESTION
4	Illustrate the working of spanning tree based termination detection algorithm	4	MODEL QUESTION
5	Specify the issues in recording a global state.	3	MODEL QUESTION
6	Explain the rules used to update clocks in scalar time representation..	3	MODEL QUESTION
7	Apply ring-based leader election algorithm with 10 processes in the worst-performing case. Count the number of messages needed.	7	MODEL QUESTION
8	Apply spanning tree-based termination detection algorithm in the following scenario. The nodes are processes 0 to 6. Leaf nodes 3, 4, 5, and 6 are each given tokens T3, T4, T5 and T6 respectively. Leaf nodes 3, 4, 5 and 6 terminate in the order, but before terminating node 5,it sends a message to node 1.	7	MODEL QUESTION
9	Explain Termination detection by weight throwing.	5	MODEL QUESTION
10	Explain Spanning-tree-based algorithm	5	MODEL QUESTION

MODULE 3			
Sl.No	**Questions**	**Marks**	**KTU/KU Month/Year**
1	Describe how quorum-based mutual exclusion algorithms differ from the other categories of mutual exclusion algorithms.	3	MODEL QUESTION PAPER

2	Explain with example, how wait-for-graphs can be used in deadlock detection.	3	MODEL QUESTION PAPER
3	Explain and illustrate Lamport's mutual exclusion algorithm.	8	MODEL QUESTION PAPER
4	Discuss the three types of messages required for deadlock handling in Maekawa's algorithm. Explain how Maekawa's algorithm handles deadlocks.	6	MODEL QUESTION PAPER
5	Explain and illustrate Ricart- Agrawala algorithm for achieving mutual exclusion.	8	MODEL QUESTION PAPER
6	Explain any three different models and issues of deadlock	6	MODEL QUESTION PAPER
7	What are the requirements of mutual exclusion algorithms?	5	MODEL QUESTION
8	Illustrate Suzuki- Kasami's broadcast algorithm.	7	MODEL QUESTION
9	Explain deadlock handling strategies	5	MODEL QUESTION

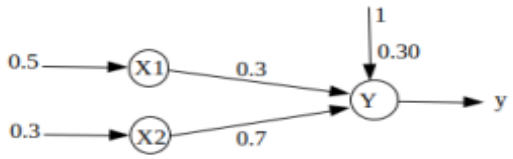
MODULE 4			
Sl. No	**Questions**	**Marks**	**KTU/KU Month/Year**
1	What are the issues in failure recovery? Illustrate with suitable examples.	7	DEC 18
2	Show that Lamport's Bakery algorithm for shared memory mutual exclusion, satisfy the three requirements of critical section problem.	7	DEC 18
3	Differentiate consistent and inconsistent states with examples.	4	DEC18
4	What is check point-based rollback-recovery? Explain the three classifications of check point-based rollback-recovery	10	DEC 18
5	Define the no-orphans consistency condition.	3	MAY 19

6	List any three advantages of using distributed shared memory.	3	MAY 19
7	Illustrate the detailed abstraction of distributed shared memory and interaction with application processes.	5	MAY 19
8	Classify different log based roll back recovery techniques.	5	MAY 19
9	Explain Lamport's bakery algorithm	7	MAR 2020
10	Describe log based roll back recovery.	7	MAR 2020

MODULE 5

Sl.No	Questions	Marks	KTU/KU Month/Year
1	Define Byzantine agreement problem.	3	MODEL QUESTION
2	What do you mean by Vice and Venus in AFS?. What are their roles?	6	DEC 18
3	Summarize any five Distributed File System requirements.	6	DEC 18
4	Explain NFS Architecture with diagram	6	DEC 18
5	Differentiate Andrew file system and NFS	4	MAY 19
6	How does Andrew File System ensure that the cached copies of files are up-to-date?	6	MAY 19
7	Briefly explain different types of navigations used for name resolution	6	MAY 19
8	With appropriate diagram explain the distribution of processes in the Andrew File System	4	MAR 2020
9	Differentiate between whole file serving and whole file caching in Andrew file system	3	MODEL QUESTION
10	Differentiate Andrew file system and NFS.	4	MODEL QUESTION
11	Explain Sun NFS architecture with diagram.	10	MODEL QUESTION
12	Explain consensus algorithm for crash failures under synchronous systems.	6	MODEL QUESTION
13	Summarize distributed file system requirements	8	MODEL QUESTION

CST 444 SOFT COMPUTING

Module I			
Sl. No	Questions	Marks	Year
1	Differentiate between the problem solving strategies: Soft computing and Hard computing.?	3	KTU (S,FE) 2021
2	What is the role of activation function in Artificial Neural Network (ANN)? Write and explain any two activation functions of ANN.	3	KTU(S,FE)2021
3	Implement AND function using Mc Culloch-Pitts neuron. Use binary data.	6	KTU(S,FE)2021
4	Calculate the net input to the neuron Y for the network shown in figure. 	5	KTU(S,FE) 2021
5	Explain the five basic architectures of ANN based on connection.	4	KTU(S,FE) 2021
6	"The Hebb rule is more suited for Bipolar data than binary data." Justify the statement.?	3	DEC 2021
7	Explain the training algorithm of Hebb network.	3	KTU (S,FE) JAN 2022
8	What are the different types of learning methods employed in neural networks?	4	KTU (S,FE) JAN 2022
9	Implementation of logic gates using i) MP neuron ii) Hebb network --AND,OR,NOR,XOR ,AND, NOT etc.	8	MODEL QUESTION
10	Obtain the output of the neuron for a network with inputs are given as [x1, x2] = [0.7, 0.8] and the weights are [w1, w2] = [0.2, 0.3] with bias = 0.9. Use i) Binary sigmoidal activation function ii) Bipolar sigmoid activation function	8	MODEL QUESTION
11	Design a Hebb net to implement NOR function using with bipolar inputs and targets.	6	KTU (R,S) DEC 2020

12	A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. Predict the output?	6	MODEL QUESTION
13	Implement AND(x1, x2) where x1, x2 is an element of {0,1} using MP neuron.	4	MODEL QUESTION
	Module II		
1	What are the activations used in backpropagation network algorithm	3	KTU (PT)DEC 2021
2	State the testing algorithm used in perceptron algorithm.	4	KTU (PT)DEC 2021
3	How is error propagated in backpropagation network? Explain the phase II of BPN training algorithm.	5	KTU (S,Fe)Jan 2022
4	Implement OR function using perceptron training algorithm with binary inputs and bipolar targets. (up to 2 epochs)	4	KTU Dec 2020
5	Draw the architecture of back propagation network and write training algorithm.	6	KTU Dec 2020
6	Explain the architecture and training algorithm of Adaline network . Use Adaline nerwork to train NOR logic function with bipolar inputs and targets. Perform 2 epochs of training.	8	Model Question
7	Discuss the training algorithm and explain the weight updates in back propagation networks	10	Model Question
8	Explain how synaptic weights are adapted iteration by iteration using error correction rule in Perceptron convergence algorithm for an OR gate with bipolar inputs and outputs. Initial weights are all zero and learning rate parameter η=0.1	10	Model Question
9	Explain Perceptron convergence theorem and discuss Perceptron algorithm based on XOR logic function.	4	Model Question
10	Implement OR function using Adaline network. Use bipolar inputs. Perform one epoch of training.	6	Model Question
	Module III		

1	Define (i) core (ii) support (iii) boundary of a fuzzy set.	5	KTU DEC 2020
2	List the properties of fuzzy set.	5	KTU DEC 2020
3	Differentiate between fuzzy tolerance and equivalence relation. How can we convert fuzzy tolerance relation to fuzzy equivalence relation?	3	KTU Dec 2020
4	Using your own intuition and your own definitions of the universe of discourse, plot fuzzy membership functions for "Age of people": (i) Very young (ii) Young (iii) Middle-aged (iv) Old (v) Very old	3	KTU Dec 2020
5	Define composition operation in fuzzy relations? Two fuzzy relations R and S are given below. Obtain the composition T of these fuzzy relations using $\underset{\sim}{R} = \begin{array}{c c} & \begin{array}{cc} y1 & y2 \end{array} \\ \begin{array}{c} x1 \\ x2 \end{array} & \begin{bmatrix} .5 & .2 \\ .3 & .8 \end{bmatrix} \end{array} \quad \underset{\sim}{S} = \begin{array}{c c} & \begin{array}{ccc} z1 & z2 & z3 \end{array} \\ \begin{array}{c} y1 \\ y2 \end{array} & \begin{bmatrix} 1 & .5 & .2 \\ .8 & .4 & .6 \end{bmatrix} \end{array}$ i)Max-min composition ii)Max-product composition	9	KTU (S,FE) 2022
6	Given two fuzzy sets A and B, compute the following set operations on A and B: a) Algebraic sum b) Algebraic product c) Bounded sum d) Bounded difference $\underset{\sim}{A} = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\}$ $\underset{\sim}{B} = \left\{ \frac{0.5}{2} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}$	5	KTU (S,FE) 2022
	Module IV		
1	Apply Mamdani fuzzy model to design a controller to determine the wash time of a domestic washing machine. Assume input is dirt and grease of the cloth. Use three descriptors for input variable and five descriptors for output variables .Derive the set of rules for controller action and defuzzification. Design should be supported by figure wherever possible.	10	Model Question

2	Using inference approach, find the membership values for each of the triangular shapes (I, R, IR, T) for a triangle with angles 120°, 50°, 10°.	8	Model Question
3	Explain Single-Point Crossover and Two-Point Crossover with example.	4	Model question
4	Differentiate between value encoding and permutation encoding.	8	Model Question
5	Explain the stopping conditions for genetic algorithm.	6	Model question
6	Explain five different types of selection and crossover genetic operators.	10	KTU(S,FE)JAN2022
7	With block diagram explain about Fuzzy Inference system.	5	KTU DEC2020
8	With an example explain Mamdani FIS model.	6	KTU(PT)DEC 2021
9	List out the various encoding techniques in GA with examples for each	5	KTU(PT)DEC 2021
MODULE V			
1	Explain convex and non convex MOOP? How to find a non dominated set.	10	Model Question
2	What are the properties of dominance relation?	4	Model Question
3	Explain Genetic Neuro-Hybrid System with block diagram. Also write the advantages of Genetic- Neuro Hybrid systems	8	Model Question
4	Discuss the classification of Neuro-Fuzzy Hybrid System.	6	Model Question
5	Give the properties of Genetic-neuro hybrid systems with a block diagram.	7	Model question
6	With suitable example,explain about different methods of encoding that are possible in genetic algorithm.	5	Model Question
7	List the various advantages of Genetic-neuro hybrid systems.	3	Model Question
8	Explain cooperative Neural Fuzzy systems.	5	Model Question
9	Explain a simple genetic algorithm with an example	5	Model Question